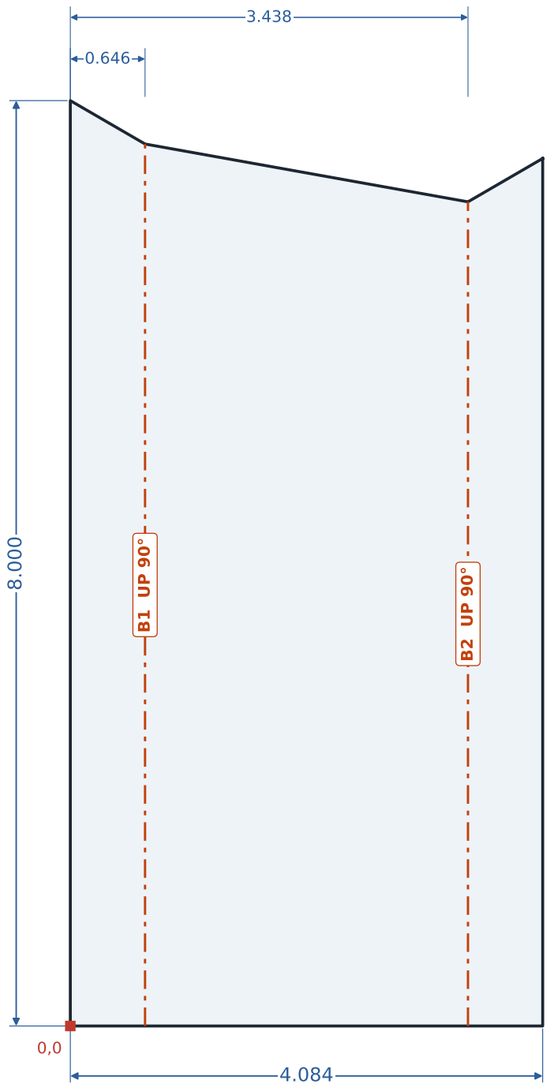
FR-3 · Channel — flat pattern (2 bends 90°)

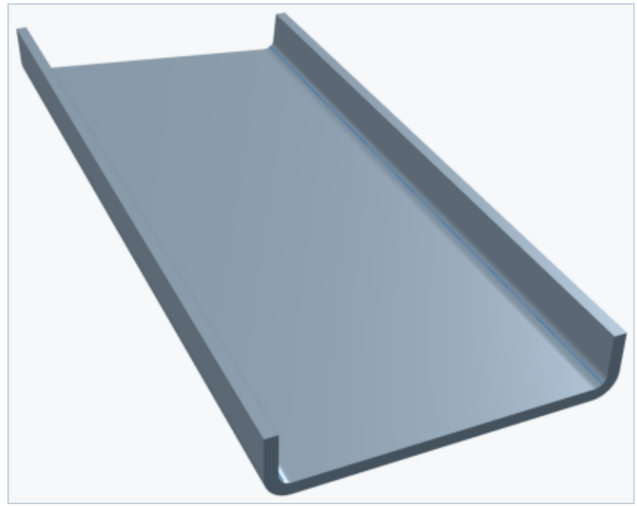
11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 4.084″ × 8.000″ · ≈ 1.03 lb each · QTY 1 FR + 1 FL = 2 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	X = 0.646
B2	90.0°	UP (toward)	R=t 0.120	2	X = 3.438

FILES

- FR-3_FR_cut.dxf (right, qty 1 / vehicle)
- FR-3_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-3_ref.dxf (reference: + bend lines, drill-later)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* bend 90° on dashed lines, 3/4 flanges inward

FR-3 Channel	HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV A**
FR-3 · Channel — flat pattern (2 bends 90°)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-03	**SHEET 1/1**